INT374 – DATA ANALYTICS WITH POWER BI PROJECT REPORT

(Project Semester January-April 2025)

Terrorism Insights & Analysis

Submitted by

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CERTIFICATE

This is to certify that Sanjivini Antil bearing Registration no.12310205 has completed INT374 project titled, "Terrorism Insights & Analysis" under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor Designation of the Supervisor

School of Computer Science and Engineering

Lovely Professional University Phagwara, Punjab.

Date: 11/12/25

DECLARATION

I, Sanjivini Antil student of Computer Science and Engineering under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 11/12/25 Signature

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ACKNOWLEDGEMENT

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CHAPTER 1: INTRODUCTION

Terrorism has emerged as one of the most severe challenges faced by modern societies across the globe. It not only results in the loss of innocent lives but also causes long-term psychological, social, and economic impacts. Governments, security agencies, and policymakers continuously seek reliable methods to analyze terrorism-related data in order to identify trends, hotspots, and patterns that can assist in prevention and preparedness.

With the rapid growth of data analytics and business intelligence tools, large volumes of historical data can now be analyzed efficiently. **Microsoft Power BI** is one such powerful visualization and analytics tool that allows users to transform raw data into meaningful insights through interactive dashboards and reports.

This project, titled **"Terrorism Insights & Analysis Dashboard"**, aims to analyze global terrorism data using Power BI. The project is divided into two major analytical perspectives:

1. **Global Terrorism Analysis** – providing a worldwide overview of terrorism incidents.
2. **Terrorism in India** – offering a focused, in-depth analysis of terrorism incidents occurring within India.

The dashboards created in this project help users understand how terrorism has evolved over time, which regions are most affected, what types of attacks are most common, and how casualties vary across regions.

LinkedIn: https://www.linkedin.com/posts/sanjivini-antil-9abaa4297_powerbi-dataanalytics-datavisualization-activity-7406316461176299520-rPs6?utm_source=social_share_video_v2&utm_medium=android_app&rcm=ACoAAEfi8pMBHctra_B27qI58QA0jdeXly3uuk0&utm_campaign=copy_link

GitHub: https://github.com/Sanjivini-26/Global-Terrosim-Dashboard

CHAPTER 2: OBJECTIVES OF THE PROJECT (EXPANDED)

The primary objectives of this project are as follows:

- To analyze historical terrorism data using Power BI
- To study global trends in terrorism incidents over different time periods
- To identify countries and regions most affected by terrorist activities
- To analyze different attack types, weapon types, and target categories
- To evaluate the severity of attacks based on casualties (killed and wounded)
- To provide a detailed country-level analysis for India
- To design interactive dashboards that allow users to filter and explore data easily
- To demonstrate practical application of data analytics concepts learned in the course

These objectives ensure that the project is not only technically sound but also socially relevant and analytically meaningful.

CHAPTER 3: Overview of the Dataset

The dataset used in this project contains detailed historical records of terrorist incidents occurring across different countries and regions of the world. Each record in the dataset represents a **single terrorism incident** and captures multiple dimensions such as time, location, attack characteristics, and impact severity.

The dataset is well-structured and suitable for **exploratory and descriptive data analysis**. It enables analysis across various perspectives including **temporal trends**, **geographical distribution**, **attack patterns**, **terrorist group activity**, and **human impact in terms of casualties**.

The presence of **latitude and longitude** information allows for advanced **geospatial analysis** using map-based visualizations in Power BI. Additionally, categorical attributes such as attack type, weapon type, and target type support comparative analysis and pattern identification.

This dataset is particularly effective for building **interactive dashboards**, as it supports filtering, slicing, aggregation, and drill-down operations across multiple dimensions.

3.3 Description of Dataset Columns

The dataset consists of the following major columns, which can be categorized into **temporal, geographical, attack-related, and impact-related attributes**:

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1. Temporal Attributes

- **Date**
  Represents the exact date on which the terrorist incident occurred.
  Used for time-based analysis, trends, and year-wise or month-wise comparisons.
- **Year**
  Extracted from the Date column.
  Used extensively in line charts, slicers, and trend analysis.
- **Month / Month Name**
  Indicates the month of occurrence.
  Helps in identifying seasonal patterns in terrorist activities.

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2. Geographical Attributes

- **Country**
  Specifies the country where the incident took place.
  Used for country-level analysis, maps, and regional comparisons.
- **Region**
  Represents the broader geographical region (e.g., South Asia, Middle East).
  Useful for high-level regional analysis.

- **State / Province**
  Indicates the state or province within a country (where available).
  Used extensively in the India-specific dashboard.
- **City**
  Identifies the city or town where the incident occurred.
  Supports detailed location-level analysis.
- **Latitude**
  Geographic latitude coordinate of the incident location.
  Used in map and bubble map visualizations.
- **Longitude**
  Geographic longitude coordinate of the incident location.
  Used along with latitude for accurate geospatial plotting.

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3. Attack-Related Attributes

- **Attack Type**
  Describes the nature of the terrorist attack, such as armed assault, bombing/explosion, assassination, or hostage taking.
  Used to analyze patterns in attack methods.
- **Target Type**
  Indicates the type of target attacked, such as civilians, military, police, or government institutions.
  Helps in understanding which groups are most affected.
- **Weapon Type**
  Specifies the weapon used during the attack, such as firearms, explosives, or vehicles.
  Useful for weapon usage analysis.
- **Terrorist Group**
  Represents the name of the terrorist organization involved in the incident (if identified).
  Used to identify the most active groups and their operational patterns.

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4. Impact and Severity Attributes

- **Killed**
  Indicates the number of people killed in the incident.
  Used to measure the fatal impact of attacks.
- **Wounded**
  Indicates the number of people injured in the incident.
  Helps assess non-fatal human impact.
- **Casualties** *(Calculated Column)*
  Represents the total number of affected people, calculated as:

Casualties = Killed + Wounded
Used as a key severity indicator in analysis and visualizations.

3.4 Dataset Structure Summary

- Each **row** represents one terrorist incident
- Each **column** represents a specific attribute of that incident
- The dataset contains both **categorical** and **numerical** variables
- The structure supports **star schema modeling** for efficient analysis

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3.5 Suitability of the Dataset

This dataset is highly suitable for:

- Trend analysis over time
- Country-wise and region-wise comparisons
- Geospatial mapping
- Attack pattern identification
- Impact assessment based on casualties

CHAPTER 4: DATA PREPROCESSING

Data preprocessing is one of the most critical stages in any data analytics project, as the quality of insights derived from a dashboard is directly dependent on the quality of the underlying data. Real-world datasets, especially those related to complex and sensitive domains such as terrorism, often contain missing values, inconsistencies, redundant attributes, and formatting issues. Therefore, it is essential to clean, transform, and structure the dataset before performing any analytical or visualization tasks.

In this project, data preprocessing was carried out using **Power Query Editor** in **Microsoft Power BI**. Power Query provides a flexible and efficient environment for cleaning, transforming, and shaping raw data into a structured format suitable for analysis and visualization. The preprocessing steps applied ensured data accuracy, consistency, improved performance, and seamless interactivity within the dashboard.

4.1 Data Collection Method

The terrorism dataset used in this project was obtained from a reliable and publicly available data source containing historical records of terrorist incidents across the world. The data was collected in a structured, tabular format where each row represented a single terrorist incident and each column described a specific attribute related to that incident.

The dataset was imported directly into Power BI in CSV format. During the initial inspection phase, the dataset was examined to understand its structure, size, attribute types, and completeness. This preliminary review helped identify data quality issues such as missing values, duplicate records, inconsistent naming conventions, and invalid geographical coordinates.

4.2 Data Cleaning Process

Data cleaning involved identifying and correcting errors, inconsistencies, and inaccuracies present in the raw dataset. Since the dataset contained a large number of records collected over different time periods and regions, cleaning was essential to ensure reliability of analysis.

4.2.1 Removal of Irrelevant and Duplicate Columns

The original dataset contained several columns that were not relevant to the objectives of the project or did not contribute to meaningful analysis or visualization. Such columns increased dataset complexity and negatively impacted dashboard performance.

The following actions were taken:

- Columns not required for analysis, such as excessive textual descriptions and metadata fields, were removed.
- Redundant identifier columns that did not add analytical value were eliminated.
- Duplicate columns containing similar information were carefully reviewed and removed.

This step helped simplify the dataset and reduce memory usage, resulting in faster report rendering.

4.2.2 Handling Missing Values

Missing values were present in several attributes, particularly in numerical fields such as **Killed**, **Wounded**, and geographical coordinates. If left untreated, these missing values could lead to incorrect aggregations and misleading insights.

The following approach was adopted:

- Missing values in **Killed** and **Wounded** columns were replaced with **zero (0)**, as the absence of recorded casualties indicates no reported deaths or injuries.
- For categorical attributes such as **Attack Type**, **Target Type**, **Weapon Type**, and **Terrorist Group**, missing values were replaced with the label **"Unknown"** to maintain categorical consistency.
- Records with missing or invalid latitude and longitude values were filtered out to prevent errors in map-based visualizations.

This ensured that all calculations and visualizations were based on complete and reliable data.

4.3 Data Transformation

Data transformation involves converting data into a format suitable for analysis and visualization. This step ensures uniformity across the dataset and enables efficient aggregations.

4.3.1 Data Type Conversion

Several columns were stored in inappropriate data formats in the raw dataset. To address this, the following transformations were applied:

- **Date** fields were converted to proper Date data types.
- **Killed**, **Wounded**, and other numeric fields were converted to whole numbers.
- **Latitude** and **Longitude** were converted to decimal numbers to support accurate geospatial mapping.
- Text-based fields such as country names and attack types were converted to text format.

Correct data typing was essential for applying DAX calculations and creating accurate visualizations.

4.3.2 Standardization of Categorical Fields

Inconsistencies were observed in categorical columns such as **Country**, **Attack Type**, and **Weapon Type** due to variations in spelling, capitalization, and naming conventions.

To standardize these fields:

- Text values were trimmed to remove extra spaces.
- Inconsistent naming patterns were unified using replace operations.

- Country and region names were standardized to avoid duplication in visuals.

Standardization ensured accurate grouping and prevented incorrect category-wise aggregations.

4.4 Feature Engineering

Feature engineering was performed to derive new attributes that enhance analytical capability and improve dashboard usability.

4.4.1 Creation of Calculated Columns

Several calculated columns were created to support deeper analysis:

- **Casualties**
  This column was calculated as the sum of killed and wounded individuals:
  Casualties = Killed + Wounded
  It serves as a key indicator of attack severity.
- **Year**
  Extracted from the Date column to support year-wise trend analysis.
- **Month** and **Month Name**
  Extracted to analyze seasonal patterns in terrorism incidents.
- **Quarter**
  Derived from the Date column to support quarterly trend analysis.

These derived attributes played a vital role in trend-based and time-series analysis.

4.5 Handling Geographical Data

Geographical attributes required special attention as they directly affected map-based visualizations:

- Rows with missing, zero, or invalid **Latitude** and **Longitude** values were removed.
- Coordinates were verified to ensure accurate plotting on maps.
- Country fields were categorized as geographical data types within Power BI.

This step ensured accurate rendering of global and India-specific maps.

4.6 Final Dataset Preparation

After completing data cleaning, transformation, and feature engineering, the final dataset was thoroughly validated:

- Sample aggregations were tested to ensure correctness.
- Visual previews were checked for consistency.

CHAPTER 5: DATA MODELING

Data modeling is a crucial step in transforming processed data into a structured format that supports efficient analysis and visualization. In this project, a **star schema data model** was implemented to improve performance, simplify relationships, and enable effective analytical querying within Power BI.

The star schema was selected because of its simplicity, scalability, and suitability for business intelligence applications. It reduces data redundancy and allows faster aggregations, making it ideal for interactive dashboards with large datasets.

5.1 Star Schema Design

The star schema consists of a central **fact table** connected to multiple **dimension tables**. This design enables users to analyze numerical measures from different perspectives such as time, location, attack type, and terrorist group.

5.2 Fact Table

Fact_Incidents

The **Fact_Incidents** table serves as the central table in the data model. Each record in this table represents a single terrorist incident. It stores quantitative and event-level information that forms the basis of analysis.

Key attributes in the fact table include:

- Date of the incident
- Location-related details (country, state, city)
- Attack type, target type, and weapon type identifiers
- Terrorist group involved
- Number of people killed and wounded
- Total casualties

The fact table contains foreign keys that link it to the corresponding dimension tables, enabling multidimensional analysis.

5.3 Dimension Tables

Dimension tables provide descriptive context to the numerical data stored in the fact table. The following dimension tables were created:

- **Dim_Date**
  Contains time-related attributes such as date, year, month, and quarter. It enables time-based analysis and trend identification.

- **Dim_Country**
  Stores country and region information. This dimension supports geographical analysis and map-based visualizations.
- **Dim_AttackType**
  Represents different types of terrorist attacks. It allows comparison of attack methods and pattern analysis.
- **Dim_TargetType**
  Contains categories of targets such as civilians, military, and government. This helps analyze the impact on different target groups.
- **Dim_WeaponType**
  Stores weapon classifications used in terrorist incidents. It enables analysis of weapon usage patterns.
- **Dim_Group**
  Contains information about terrorist organizations involved in attacks. This dimension helps identify the most active groups.

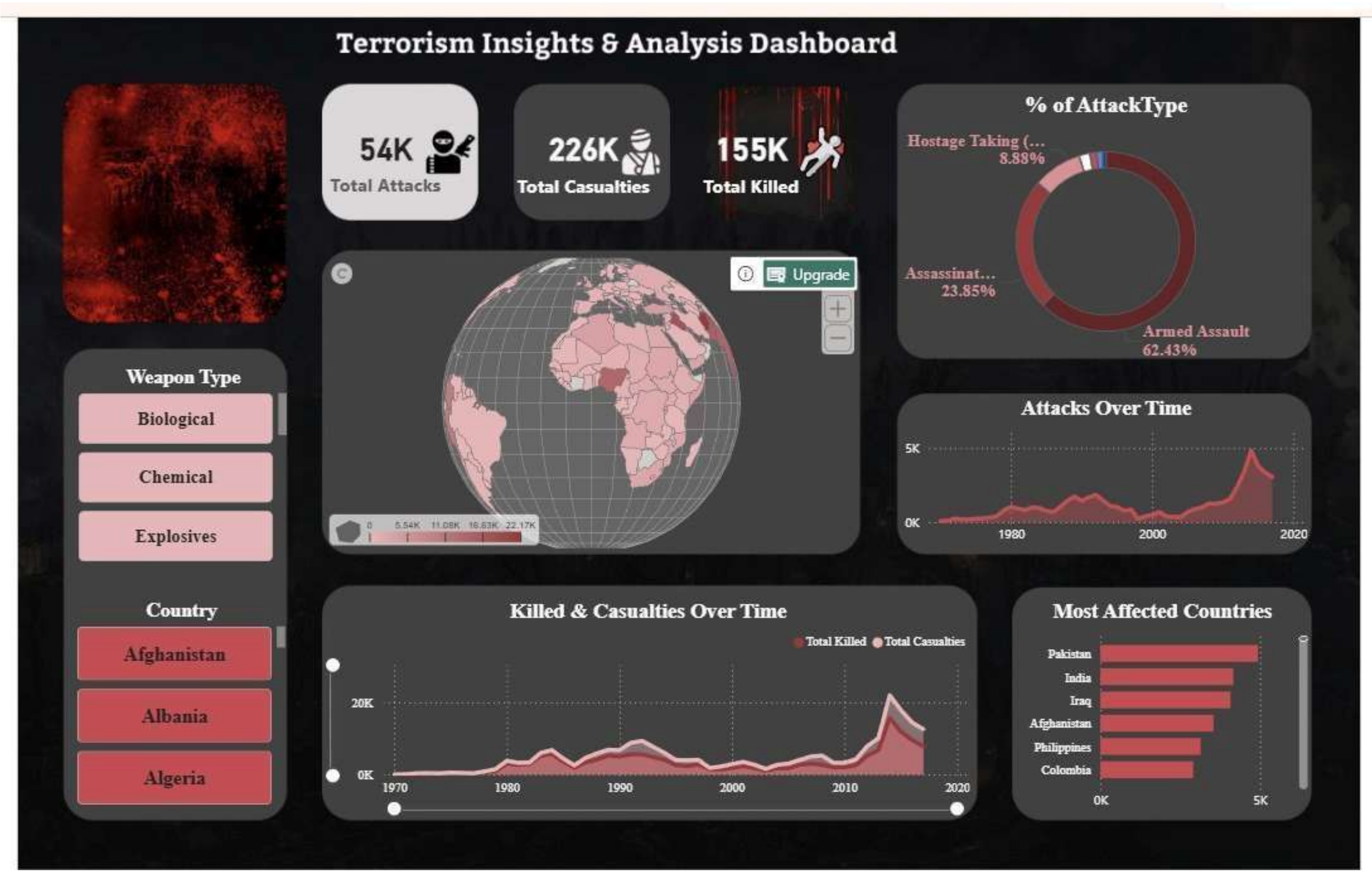
5.4 Relationships and Model Efficiency

All dimension tables were connected to the fact table using **one-to-many relationships**, with dimensions on the "one" side and the fact table on the "many" side. Single-directional filtering was used to maintain model clarity and avoid ambiguity.

This modeling approach enables efficient filtering, accurate aggregations, and smooth interactivity across all dashboard visuals. The star schema design significantly improved query performance and made the data model easy to understand and maintain.

CHAPTER 6 : DASHBOARD

6.1 Global Terrorism Dashboard



The **Global Terrorism Dashboard** provides a comprehensive overview of terrorism incidents occurring across different countries and regions worldwide. The primary objective of this dashboard is to enable users to understand global patterns, trends, and the overall impact of terrorist activities through interactive and visually intuitive components.

The dashboard is designed using a dark theme to enhance readability and emphasize critical insights. It consists of the following key visual elements:

Key Performance Indicator (KPI) Cards

KPI cards are placed at the top of the dashboard to present a quick summary of the global terrorism scenario. These cards display:

- Total number of terrorist incidents
- Total number of casualties
- Total number of people killed

These indicators provide an instant snapshot of the scale and severity of terrorism worldwide.

Global Terrorism Heat Map

A filled map visualization is used to represent the geographical distribution of terrorist incidents. Countries with a higher frequency of incidents are highlighted using darker color shades, making it easy to identify global hotspots. This visualization helps users quickly recognize regions that are most affected by terrorism.

Attacks Over Time Line Chart

The line chart displays the trend of terrorist incidents over time. By plotting the number of attacks against years, this visual helps identify periods of increase or decline in terrorism activities. It allows users to analyze long-term patterns and observe how terrorism has evolved globally.

Attack Type Distribution Donut Chart

This donut chart illustrates the proportion of different attack types such as armed assaults, bombings, assassinations, and hostage-taking incidents. It provides insight into the most commonly used attack methods and helps understand operational preferences of terrorist groups.

Most Affected Countries Bar Chart

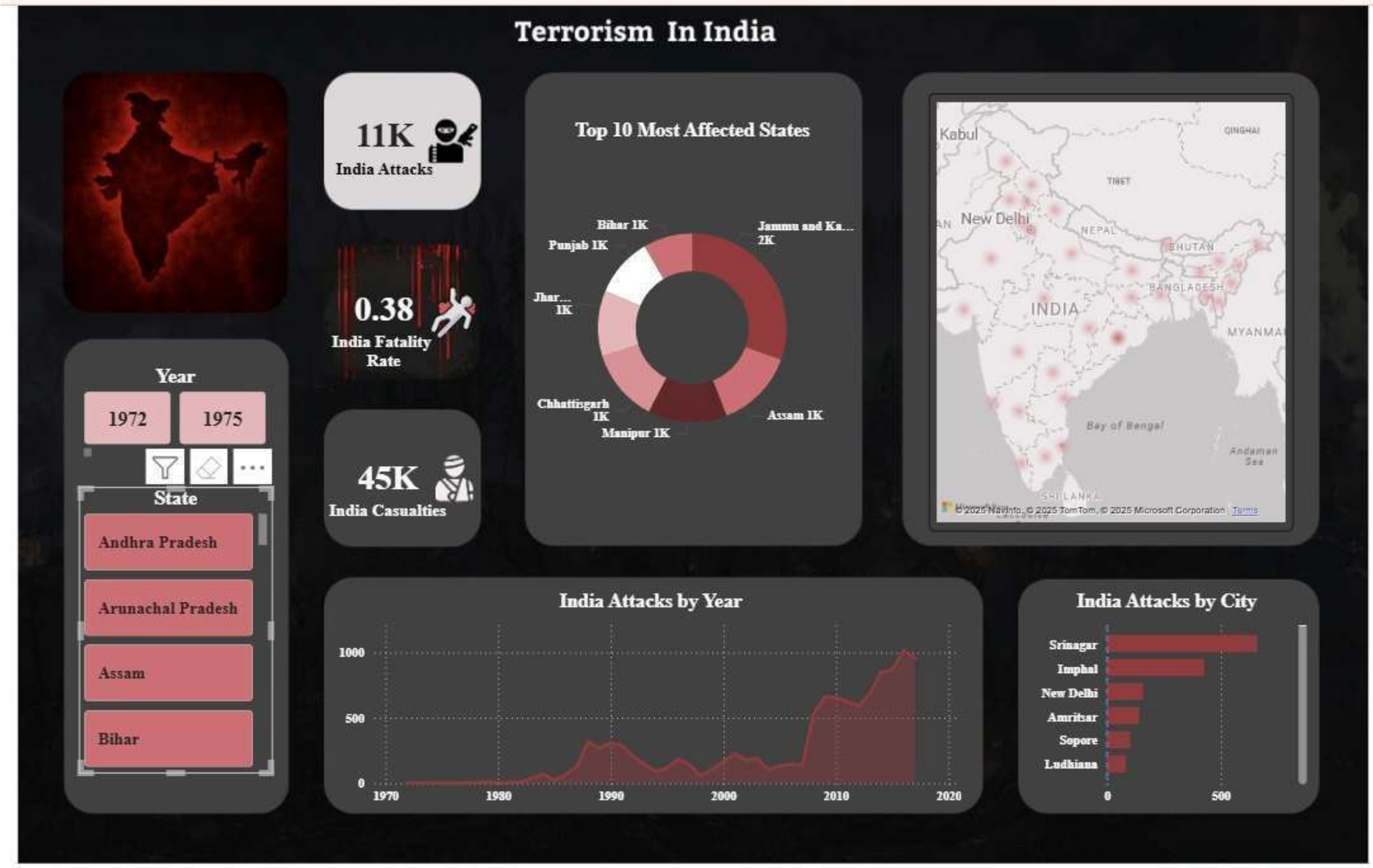
A horizontal bar chart ranks countries based on the number of terrorist incidents. This visualization clearly highlights the top affected countries and supports comparative analysis across nations.

Interactive Slicers

Interactive slicers are provided for attributes such as **country** and **weapon type**. These slicers allow users to dynamically filter the dashboard and explore specific regions or attack characteristics, enhancing analytical flexibility.

Overall, the Global Terrorism Dashboard enables a high-level yet detailed understanding of terrorism patterns at a worldwide scale.

6.2 Terrorism in India Dashboard



The **Terrorism in India Dashboard** is designed to provide a focused and detailed analysis of terrorism incidents occurring specifically within India. Unlike the global dashboard, this page restricts the data scope to India using page-level filters, ensuring that all visualizations represent India-only data.

This dashboard supports region-specific insights and helps identify internal patterns and hotspots within the country.

India-Specific KPI Cards

The KPI cards on this dashboard present key statistics related to terrorism in India, including:

- Total terrorist incidents in India
- Total casualties
- Total number of people killed

These KPIs offer an immediate overview of the severity and scale of terrorism within the country.

India Terrorism Heat Map

A map visualization is used to show the distribution of terrorist incidents across different locations in India. The map highlights areas with higher incident density, enabling users to identify regions and cities that are most affected.

Top Affected States Donut Chart

This donut chart displays the distribution of terrorist incidents across the most affected states in India. It provides a clear comparison of state-wise impact and helps identify regional concentration of terrorist activities.

Attacks by Year Line Chart

The line chart illustrates year-wise trends in terrorist incidents within India. This visual helps analyze how terrorism has changed over time at the national level and identifies periods of increased or reduced activity.

Attacks by City Bar Chart

A bar chart is used to rank cities based on the number of terrorist incidents. This allows for detailed city-level analysis and helps identify urban centers that experience higher terrorist activity.

Interactive Slicers

Slicers for **year** and **state** are included to allow users to filter data dynamically. These filters enhance interactivity and enable focused analysis for specific time periods or regions within India.

In summary, the India dashboard complements the global dashboard by providing a deeper, country-specific perspective. Together, both dashboards offer a comprehensive and layered understanding of terrorism patterns.

CHAPTER 7: ANALYSIS AND INSIGHTS

This chapter presents the key findings and insights derived from the **Global Terrorism Dashboard** and the **Terrorism in India Dashboard**. The analysis is based on interactive visualizations such as KPI cards, maps, line charts, bar charts, and donut charts, which collectively help in identifying patterns, trends, and high-risk areas.

7.1 Global Terrorism Analysis

The global dashboard reveals that terrorism incidents are not evenly distributed across the world. Certain regions and countries experience significantly higher levels of terrorist activity compared to others. The KPI indicators highlight the overall magnitude of global terrorism in terms of total incidents, casualties, and fatalities.

The **global heat map** clearly identifies terrorism hotspots, indicating that regions such as South Asia and the Middle East are heavily affected. Countries within these regions show a higher concentration of incidents, reflecting prolonged conflict and instability.

7.2 Trend Analysis Over the Years

The line chart depicting terrorist incidents over time shows noticeable fluctuations across different years. Certain periods exhibit sharp increases in incidents, while others show a decline. These variations suggest that terrorism is influenced by geopolitical events, regional conflicts, and counterterrorism measures implemented during specific periods.

The trend analysis also highlights that terrorism is not static but evolves over time, making continuous monitoring and analysis essential.

7.3 Attack Type and Weapon Analysis

Analysis of attack types indicates that **armed assaults and bombings/explosions** are among the most commonly used methods by terrorist organizations. The attack type distribution chart demonstrates a clear preference for methods that can cause widespread damage and casualties.

Weapon type analysis further supports this observation, as explosives and firearms account for a major share of incidents. These findings emphasize the need for focused security measures to counter these specific attack methods.

7.4 Country-Wise Impact Analysis

The bar chart showing the most affected countries ranks nations based on the number of terrorist incidents. A small group of countries contributes disproportionately to the global incident count, indicating localized but intense terrorism activity.

This visualization helps in comparative analysis and assists policymakers in identifying regions that require prioritized attention and resources.

7.5 India-Specific Analysis

The India-focused dashboard provides deeper insights into terrorism within the country. The KPI cards show the overall scale of terrorist activity in India, while the heat map highlights region-wise and city-level concentration of incidents.

Certain states and regions consistently appear as hotspots, indicating long-term security challenges. The city-wise bar chart further reveals that terrorist activity is concentrated in specific urban and conflict-prone areas.

7.6 Temporal Patterns in India

The year-wise trend analysis for India shows fluctuations in terrorist activity over time. Some years reflect increased incidents, while others show relative decline, possibly due to changes in security policies and counterterrorism operations.

The ability to filter by year and state enables users to perform detailed temporal analysis and understand how terrorism patterns have evolved within India.

7.7 Key Observations

The major observations from the analysis include:

- Terrorism incidents are highly region-specific rather than uniformly distributed
- Certain countries and regions act as persistent hotspots
- Armed assaults and bombings are the most common attack types
- In India, terrorism is concentrated in specific states and cities
- Casualties vary significantly depending on attack method and location

These insights demonstrate the effectiveness of interactive dashboards in uncovering meaningful patterns from complex datasets.

CHAPTER 8: CONCLUSION

This project successfully demonstrates the application of **Power BI** for analyzing and visualizing terrorism data. By transforming raw historical data into interactive dashboards, the project enables users to explore global and country-specific terrorism patterns efficiently.

The use of a **star schema data model**, effective preprocessing techniques, and well-designed visualizations ensured accurate analysis and smooth performance. The dual-dashboard approach provided both a global overview and a focused analysis of terrorism in India, making the study comprehensive and insightful.

Overall, the project meets all the defined objectives and highlights the importance of data analytics and visualization in understanding complex security-related issues.

CHAPTER 9: FUTURE SCOPE

Although the current project successfully analyzes historical terrorism data and presents meaningful insights through interactive Power BI dashboards, there is considerable scope for further enhancement. With the rapid advancement of data analytics, machine learning, and real-time data systems, this project can be extended in several directions to increase its analytical depth, accuracy, and real-world applicability.

9.1 Integration of Real-Time Data

One major enhancement would be the integration of **real-time or near real-time terrorism data** from live data sources such as government portals, news APIs, or security databases. Real-time data integration would allow continuous monitoring of terrorism incidents and enable dashboards to reflect the latest developments. This enhancement would be particularly useful for security agencies and policymakers who require up-to-date information for timely decision-making.

9.2 Predictive Analytics and Forecasting

The current project focuses on descriptive analysis of past data. In the future, **predictive analytics techniques** can be applied to forecast possible trends in terrorism incidents. Machine learning models such as time-series forecasting or classification algorithms could be used to predict high-risk regions, potential attack periods, or severity levels. This would transform the dashboard from a purely analytical tool into a **decision-support system**.

9.3 Inclusion of Socio-Economic and Political Indicators

To gain deeper insights into the causes and patterns of terrorism, the dataset can be enriched with **socio-economic and political indicators** such as unemployment rates, literacy levels, political instability indices, and economic growth data. Combining these factors with terrorism data would enable correlation analysis and help identify underlying drivers of terrorist activities.

9.4 Advanced Geospatial Analysis

Future versions of the project can include **advanced geospatial techniques** such as heat intensity mapping, clustering, and spatial trend analysis. Techniques like hotspot clustering can help identify emerging regions of concern and visualize changes in terrorism intensity over time. This would enhance geographical insights and improve the accuracy of spatial analysis.

9.5 Comparative and Cross-Country Analysis

The dashboard can be extended to support **comparative analysis across multiple countries or regions**. Users could compare terrorism trends between neighboring countries or across different regions to identify similarities and differences in attack patterns. Such analysis would be useful for regional security cooperation and strategic planning.

9.6 Dashboard and User Experience Enhancements

Additional improvements can be made to enhance user experience by incorporating features such as drill-through reports, advanced tooltips, and personalized dashboards. These features would allow users to explore data at multiple levels of detail and gain customized insights based on their analytical needs.

9.7 Integration with Other Analytical Tools

The project can also be integrated with other analytical platforms such as Python or R for advanced statistical analysis and modeling. This integration would further expand the analytical capabilities of the system and support more complex research objectives.

9.8 Summary of Future Scope

In summary, the future enhancements outlined above have the potential to significantly strengthen the analytical capabilities and practical relevance of the project. By incorporating real-time data, predictive analytics, advanced geospatial techniques, and enriched datasets, the dashboard can evolve into a comprehensive and intelligent analytics platform for terrorism analysis.